The Multiplicity of Meaning, Appendix B

CIAN DORR

16th June 2025

1 *Review: the theory of co-expressing*

2 *Self-reflexive syntax*

We want a language that can talk about its own syntax (or at least, the syntax of its applicative fragment). One way to get that is to add to the general syntactic theory from Appendix A new constants that make us making terms that stand for the constants of that theory, and for the new constants:

$$\langle\epsilon\rangle : \kappa \qquad \langle\kappa\rangle : \kappa \qquad \langle e\rangle : \kappa \qquad \langle t\rangle : \kappa$$
$$\langle\rightarrow\rangle : \epsilon \qquad \langle\forall\rangle : \kappa \rightarrow \epsilon$$
$$\langle\mathsf{S}\rangle : \kappa \rightarrow \kappa \rightarrow \kappa \rightarrow \epsilon \qquad \langle\mathsf{K}\rangle : \kappa \rightarrow \kappa \rightarrow \epsilon \qquad \langle\mathsf{I}\rangle : \kappa \rightarrow \epsilon$$
$$\langle\circ\rangle : \epsilon \qquad \langle\Rightarrow\rangle : \epsilon \qquad \langle\text{Const}\rangle : \epsilon \qquad \langle\text{NBase}\rangle : \epsilon \qquad \langle\text{TBase}\rangle : \epsilon$$
$$\langle\mathsf{T}\rangle : \kappa \rightarrow \epsilon \qquad \langle\mathsf{E}\rangle : \kappa \rightarrow \epsilon$$
$$\text{Clab} : \epsilon \rightarrow \epsilon \qquad \langle\text{Clab}\rangle : \epsilon$$

We add some boring axioms specifying the types of all the constants.

As a metametalinguistic abbreviation, add a quotation-forming operation $\ulcorner\cdot\urcorner$. $\ulcorner\forall_t\neg\urcorner$ is short for $\langle\forall\rangle\langle t\rangle\circ\langle\neg\rangle$. Underlined type-$\epsilon$ variables are 'anti-quoted': $\ulcorner\forall_t\underline{a}\urcorner$ is short for $\langle\forall\rangle\langle t\rangle\circ a$.

We can define an object-language expression Label $: \epsilon \rightarrow \epsilon$ that implements the definition of $\ulcorner\cdot\urcorner$, and prove every instance of $\text{Label}(\ulcorner A\urcorner) = \ulcorner\ulcorner A\urcorner\urcorner$. We can thus prove the diagonal lemma: for any $H : \epsilon \rightarrow t$, $\mathsf{Syn+}$ proves:

$$\begin{aligned}
&(\lambda a.H(a\circ \text{Label(a)}))\ulcorner(\lambda a.H(a\circ \text{Label(a)}))\urcorner \\
&\qquad\leftrightarrow H(\ulcorner(\lambda a.H(a\circ \text{Label(a)}))\urcorner\circ \text{Label}(\ulcorner(\lambda a.H(a\circ \text{Label(a)}))\urcorner)) \\
&\qquad\leftrightarrow H(\ulcorner(\lambda a.H(a\circ \text{Label(a)}))\urcorner\circ\ulcorner\ulcorner(\lambda a.H(a\circ \text{Label(a)}))\urcorner\urcorner) \\
&\qquad\leftrightarrow H(\ulcorner(\lambda a.H(a\circ \text{Label(a)}))\ulcorner(\lambda a.H(a\circ \text{Label(a)}))\urcorner\urcorner)
\end{aligned}$$

3 *Self-reflexive semantics*

DE $\qquad \ulcorner A_1\urcorner, ..., \ulcorner A_n\urcorner \ \mathsf{E}_{\sigma_1,...,\sigma_n} \ A_1, ..., A_n$

where $A_1, ..., A_n$ may be replaced by any closed terms of types $\sigma_1, ..., \sigma_n$ respectively.

Uniqueness $\qquad a \mathsf{E}_\sigma x \wedge a \mathsf{E}_\sigma y \to x =_\sigma y$

Unsurprisingly, this conflicts with DE. Where Het abbreviates $\lambda y. \exists Z(y \mathsf{E}_{\epsilon \to t} Z \wedge \neg Z y)$, the DE-instance $\ulcorner \text{Het} \urcorner \mathsf{E}_{\epsilon \to t}$ Het logically implies

$$\exists Y^{\epsilon \to t} Z^{\epsilon \to t}.\ \ulcorner \text{Het} \urcorner \mathsf{E}_{\epsilon \to t} Y \wedge \ulcorner \text{Het} \urcorner \mathsf{E}_{\epsilon \to t} Z \wedge Y \neq Z$$

Let a 'safe' constant be one not of the form $\mathsf{E}_{\vec{\sigma}}$. The baseline theory C = the syntactic theory $\mathsf{Syn+}$ from last week, plus the general semantic theory Sem, plus:

Existence $\qquad \mathsf{T}_\sigma s \wedge a :: s \to \mathsf{M}_\sigma a$

Safe Constants $\qquad \ulcorner C \urcorner \mathsf{E}_\sigma x \leftrightarrow x =_\sigma C \quad$ (where C is a safe constant of type σ)

$\mathsf{CD} = \mathsf{C} + DE$; $\mathsf{CU} = \mathsf{C}$ + Uniqueness.

4 *Some facts about the baseline theory*

5 *How many meanings?*

6 *The truth theories in* C, CD, *and* CU

Truth-of, falsehood-of, weak-truth-of:

$$T_{\vec{\sigma}} := \lambda a \vec{x}.\ a :: \ulcorner \vec{\sigma} \to t \urcorner \wedge \forall Y^{\vec{\sigma} \to t}.\ a \mathsf{E}_{\vec{\sigma} \to t} Y \to Y \vec{x}$$
$$F_{\vec{\sigma}} := \lambda a \vec{x}.\ a :: \ulcorner \vec{\sigma} \to t \urcorner \wedge \forall Y^{\vec{\sigma} \to t}.\ a \mathsf{E}_{\vec{\sigma} \to t} Y \to \neg Y \vec{x}$$
$$W_{\vec{\sigma}} := \lambda a \vec{x}.\ a :: \ulcorner \vec{\sigma} \to t \urcorner \wedge \exists Y^{\vec{\sigma} \to t}.\ a \mathsf{E}_{\vec{\sigma} \to t} Y \wedge Y \vec{x}$$

- Whenever T is an extension of C and P is a closed substitution instance of a theorem P' of T, where P' involves only safe constants (but may contain free variables), $\mathsf{T} \vdash T \ulcorner P \urcorner$
- $\mathsf{C} \vdash T \ulcorner \underline{a} \to \underline{b} \urcorner \to Ta \to Tb$.

Some axioms and rules from Friedman and Sheard:

	unparameterized	*parameterized*
T-In	$P \to T\ulcorner P \urcorner$	$A\vec{x} \to T_{\vec{\sigma}}\ulcorner A \urcorner \vec{x}$
T-Out	$T\ulcorner P \urcorner \to P$	$T_{\vec{\sigma}}\ulcorner A \urcorner \vec{x} \to A\vec{x}$
T-Cons	$Ta \to Wa$	$T_{\vec{\sigma}}a\vec{x} \to W_{\vec{\sigma}}a\vec{x}$
T-Comp	$Wa \to Ta$	$W_{\vec{\sigma}}a\vec{x} \to T_{\vec{\sigma}}a\vec{x}$
U-Inf	$(\forall y^\tau.\, T_\tau ay) \to T\ulcorner \forall_\tau \underline{a} \urcorner$	$(\forall y^\tau.\, T_{\tau,\vec{\sigma}}ay\vec{x}) \to T_{\vec{\sigma}}\ulcorner \lambda\vec{x}.\forall y^\tau.\underline{a}y\vec{x} \urcorner \vec{x}$
E-Inf	$T\ulcorner \exists_\tau \underline{a} \urcorner \to \exists y^\tau.\, T_\tau ay$	$T_{\vec{\sigma}}\ulcorner \lambda\vec{x}.\exists y^\tau.\underline{a}y\vec{x} \urcorner \vec{x} \to \exists y^\tau.\, T_{\tau,\vec{\sigma}}ay\vec{x}$
T-Intro	$P \ / \ T\ulcorner P \urcorner$	$A\vec{B} \ / \ T_{\vec{\sigma}}\ulcorner A \urcorner \vec{B}$
T-Elim	$T\ulcorner P \urcorner \ / \ P$	$T_{\vec{\sigma}}\ulcorner A \urcorner \vec{B} \ / \ A\vec{B}$
¬T-Intro	$P \ / \ W\ulcorner P \urcorner$	$A\vec{B} \ / \ W_{\vec{\sigma}}\ulcorner A \urcorner \vec{B}$
¬T-Elim	$W\ulcorner P \urcorner \ / \ P$	$W_{\vec{\sigma}}\ulcorner A \urcorner \vec{B} \ / \ A\vec{B}$

- (a) C includes all instances of T-Cons (parameterized).
  (b) C includes all instances of U-Inf (parameterized).
  (c) C therefore also includes all instances of the analogue of U-Inf for universal quantification restricted by a safe predicate, i.e. $(\forall y^\tau.Hy \to T_{\tau,\vec{\sigma}}ay\vec{x}) \to T_{\vec{\sigma}}\ulcorner \lambda\vec{z}.\forall y^\tau.Hy \to \underline{a}\vec{z} \urcorner \vec{x}$.
  (d) Any theory that includes C and includes unparameterized T-In is inconsistent.
  (e) Any theory that includes C and is closed under parameterized T-Intro is inconsistent.
  (e) Any theory that includes C and is closed under unparameterized T-Intro is ω-inconsistent.
  (f) C is closed under W-Elim and T-Elim.
- T-Out (parameterized) is equivalent in Sem to DE; T-Comp and E-Inf are both equivalent in C to Uniqueness (and hence to each other).
- Closing CD under T-Intro or W-Elim would lead to inconsistency.

An unsophisticated way of getting more 'reflective':

$$\mathsf{C}_0 = \mathsf{C} \qquad \mathsf{C}_{n+1} = \mathsf{C}_n + \{T\ulcorner P \urcorner \mid \mathsf{C}_n \vdash P\}$$

$$\mathsf{CD}_n = \mathsf{C}_n + DE$$

$$\mathsf{CU}_0 = \mathsf{CU} \qquad \mathsf{CU}_{n+1} = \mathsf{CU}_n + \{T\ulcorner P \urcorner \mid \mathsf{CU}_n \vdash P\}$$

For a more sophisticated approach, define

$$\mathrm{Ref}_{\mathsf{T}} \qquad \forall a^\epsilon.\ \mathrm{Prv}_{\mathsf{T}}\, a \to Ta$$

where for recursively axiomatizable T with provability predicate $\mathrm{Prv}_{\mathsf{T}}$.

$$\mathsf{C}^+_0 = \mathsf{C} + \mathrm{Ref}_{\mathsf{Syn+}} \qquad \mathsf{C}^+_{n+1} = \mathsf{C}^+_n + \mathrm{Ref}_{\mathsf{C}_n}$$

$$\mathsf{CD}^+_n = \mathsf{C}^+_n + DE$$

$$\mathsf{CU}^+_0 = \mathsf{CU} + \mathrm{Ref}_{\mathsf{Syn+}} \qquad \mathsf{CU}^+_{n+1} = \mathsf{CU}^+_n + \mathrm{Ref}_{\mathsf{CU}_n}$$

7 *Consistency results*

Each CD^+_n and CU^+_n is consistent. (And hence their limits CD^+_ω and CU^+_ω are consistent too, though ω-inconsistent.)

8 *Iteration principles*

$$
\begin{aligned}
TT(a) &:= T_\epsilon \ulcorner T \urcorner a & \qquad TT_{\vec\sigma} a\vec{x} &:= T_{\epsilon,\vec\sigma} \ulcorner T_{\vec\sigma} \urcorner a\vec{x} \\
WW(a) &:= W_\epsilon \ulcorner W \urcorner a & \qquad WW_{\vec\sigma} a\vec{x} &:= W_{\epsilon,\vec\sigma} \ulcorner W_{\vec\sigma} \urcorner a\vec{x} \\
TW(a) &:= T_\epsilon \ulcorner W \urcorner a & \qquad TW_{\vec\sigma} a\vec{x} &:= T_{\epsilon,\vec\sigma} \ulcorner W_{\vec\sigma} \urcorner a\vec{x} \\
WT(a) &:= W_\epsilon \ulcorner T \urcorner a & \qquad WT_{\vec\sigma} a\vec{x} &:= W_{\epsilon,\vec\sigma} \ulcorner T_{\vec\sigma} \urcorner a\vec{x}
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{T} &\to \mathrm{TT}^{\mathsf{C}^1\times} & \qquad \mathrm{TT} &\to \mathrm{T}^{\mathsf{CD}\checkmark} \\
\mathrm{W} &\to \mathrm{WT} & \qquad \mathrm{WT} &\to \mathrm{W} \\
\mathrm{T} &\to \mathrm{WT}^{\mathsf{CD}\checkmark} & \qquad \mathrm{WT} &\to \mathrm{T}^{\mathsf{CD}\times} \\
\mathrm{W} &\to \mathrm{TT}^{\mathsf{C}^1\times}_{\mathsf{CD}\times} & \qquad \mathrm{TT} &\to \mathrm{W}^{\mathsf{CD}\checkmark} \\
\mathrm{WT} &\to \mathrm{TW} & \qquad \mathrm{TW} &\to \mathrm{WT} \\
\mathrm{TT} &\to \mathrm{WW}^{\mathsf{C}^1\checkmark}_{\mathsf{CD}\checkmark} & \qquad \mathrm{WW} &\to \mathrm{TT}^{\mathsf{CD}\times} \\
\mathrm{TT} &\to \mathrm{TW}^{\mathsf{C}^1\checkmark} & \qquad \mathrm{TW} &\to \mathrm{TT} \\
\mathrm{TT} &\to \mathrm{WT}^{\mathsf{C}\checkmark} & \qquad \mathrm{WT} &\to \mathrm{TT}^{\mathsf{CD}\times}
\end{aligned}
$$

Five of these are unsettled by CD^1: two plausible, three implausible.

T→TW Every relation expressed by 'express' is such that for every term, there is something it both expresses and bears that relation to:

$$\vec{a} :: \ulcorner \vec\sigma \urcorner \wedge \ulcorner \mathsf{E}_{\vec\sigma} \urcorner\ \mathsf{E}_{\epsilon^n \to \vec\sigma \to t}\ R \to \exists \vec{x}.\, \vec{a}\ \mathsf{E}_{\vec\sigma}\ \vec{x} \wedge R\vec{a}\vec{x}$$

WT→TW Any two relations expressed by 'express' are such that there is something to which any given term bears both of them.

$$\vec{a} :: \ulcorner \vec\sigma \urcorner \wedge (\mathsf{E}_{\vec\sigma})\ \mathsf{E}_{\epsilon^n \to \vec\sigma \to t}\ R \wedge (\mathsf{E}_{\vec\sigma})\ \mathsf{E}_{\epsilon^n \to \vec\sigma \to t}\ R' \to \exists \vec{x}.\, R\vec{a}\vec{x} \wedge R'\vec{a}\vec{x}$$

TW→WT For any property, if a term bears every relation expressed by 'express' to something with that property, there's a relation expressed by 'express' that it *only* bears to things with that property.

$$(\forall R. \ulcorner \mathsf{E}_{\vec\sigma} \urcorner\ \mathsf{E}_{\epsilon^n \to \vec\sigma \to t}\ R \to \exists \vec{X}.\, S\vec{x} \wedge R\vec{a}\vec{x}) \to \exists R.\, \ulcorner \mathsf{E}_{\vec\sigma} \urcorner\ \mathsf{E}_{\epsilon^n \to \vec\sigma \to t}\ R \wedge \forall \vec{x}. R\vec{a}\vec{x} \to S\vec{x}$$

W→WT For everything a term expresses, there is a relation expressed by 'express' that it doesn't bear to anything else.

$$\vec{a}\ \mathsf{E}_{\vec{\sigma}}\ \vec{x} \rightarrow \exists R.\ \ulcorner \mathsf{E}_{\vec{\sigma}} \urcorner\ \mathsf{E}_{\epsilon^n \rightarrow \vec{\sigma} \rightarrow t}\ R \wedge \forall \vec{y}.\ R\vec{a}\vec{y} \rightarrow \vec{y} =_{\vec{\sigma}} \vec{x}$$

WW→WT For everything to which a term bears a relation expressed by 'express', there is a relation expressed by 'express' that it doesn't bear to anything else.

$$\ulcorner \mathsf{E}_{\vec{\sigma}} \urcorner\ \mathsf{E}_{\epsilon^n \rightarrow \vec{\sigma} \rightarrow t}\ R \wedge R\vec{a}\vec{x} \rightarrow \exists R^{\epsilon^n \rightarrow \vec{\sigma} \rightarrow t}.\ \ulcorner \mathsf{E}_{\vec{\sigma}} \urcorner\ \mathsf{E}_{\epsilon^n \rightarrow \vec{\sigma} \rightarrow t}\ R \wedge \forall \vec{y}.\ R\vec{a}\vec{y} \rightarrow \vec{y} =_{\vec{\sigma}} \vec{x}$$

The following principle implies WT→TW (and hence T→TW) in CD:

Core Meanings $\qquad \vec{a} :: \ulcorner \vec{\sigma} \urcorner \rightarrow \exists \vec{x}.\ \forall R.\ \ulcorner \mathsf{E}_{\vec{\sigma}} \urcorner\ \mathsf{E}_{\epsilon^n \rightarrow \vec{\sigma} \rightarrow t}\ R \rightarrow R\vec{a}\vec{x}$

The following is inconsistent with TW→WT (and hence W→WT and WW→WT):

Core Multiplicity

$$\exists a^{\epsilon} x^{\sigma} y^{\sigma}.\ x \neq_{\sigma} y \wedge \forall R^{\epsilon \rightarrow \sigma \rightarrow t}.\ \ulcorner \mathsf{E}_{\sigma} \urcorner\ \mathsf{E}_{\epsilon \rightarrow \sigma \rightarrow t}\ R \rightarrow Rax \wedge Ray$$

9 *Consistency results relating to iteration principles*

The following are consistent (and ω-consistent):

(i) CD_n^+ + WW→WT

(ii) CD_n^+ + Core Meanings + TW→WT

(iii) CD_n^+ + Core Meanings + Core Multiplicity

(iv) CD_n^+ + WT→W + W→WT

(v) CD_0^+ + T→TT